

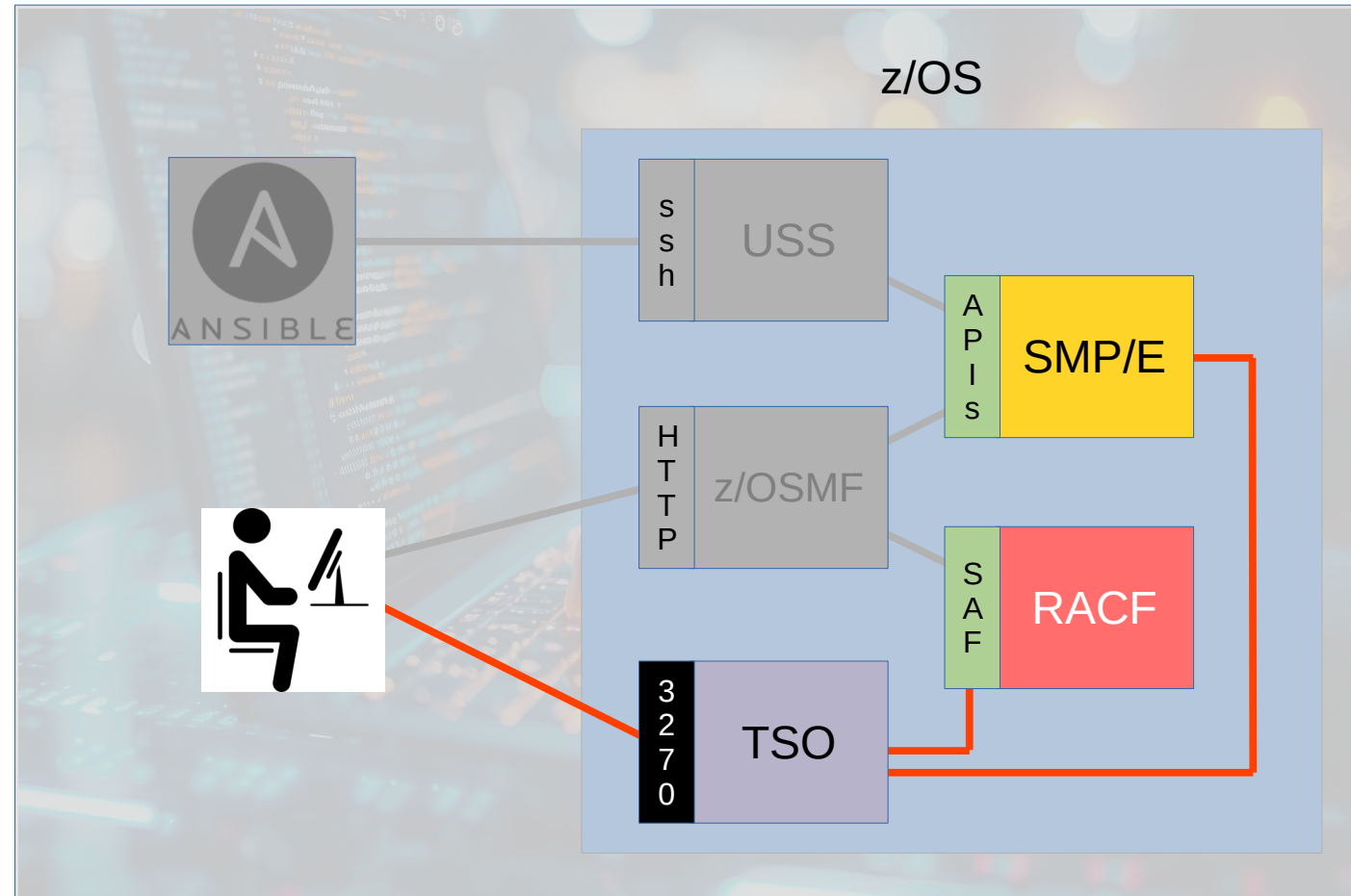
z/OSMF – the z/OS Management Facility





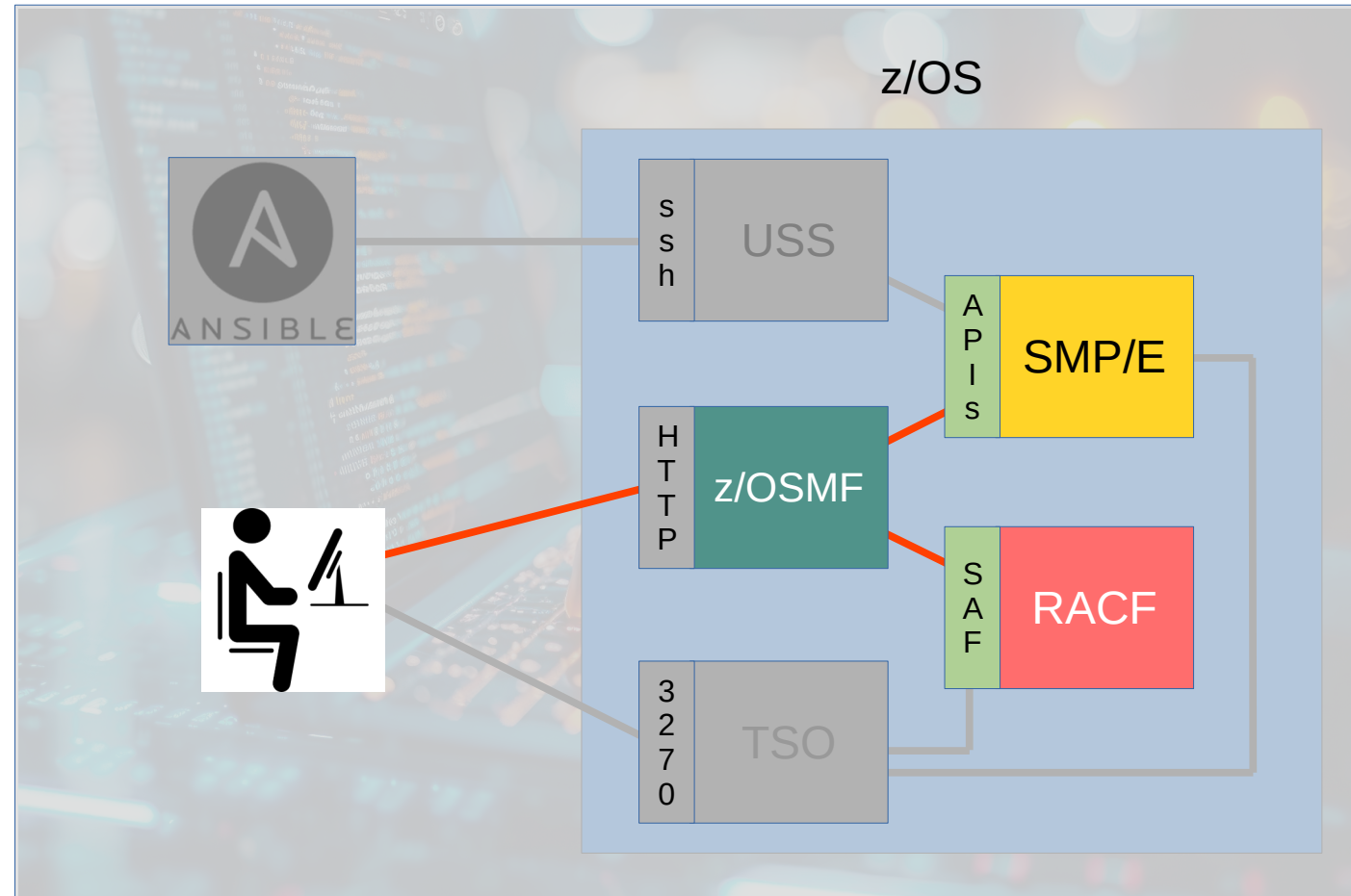
WE WERE WORKING ON THIS

Tools
Userid
Authorization
Data Sets



WE ARE WORKING ON THIS NOW

Tools
Userid
Authorization
Data Sets



SOFTWARE INSTALLATION AND MANAGEMENT

SMP/E

System Modification Program Extended

The package manager for z/OS

z/OSMF



z/OS Management Facility

Browser-based front-end to z/OS system management tools; also exposes APIs used by external tools.

Ansible

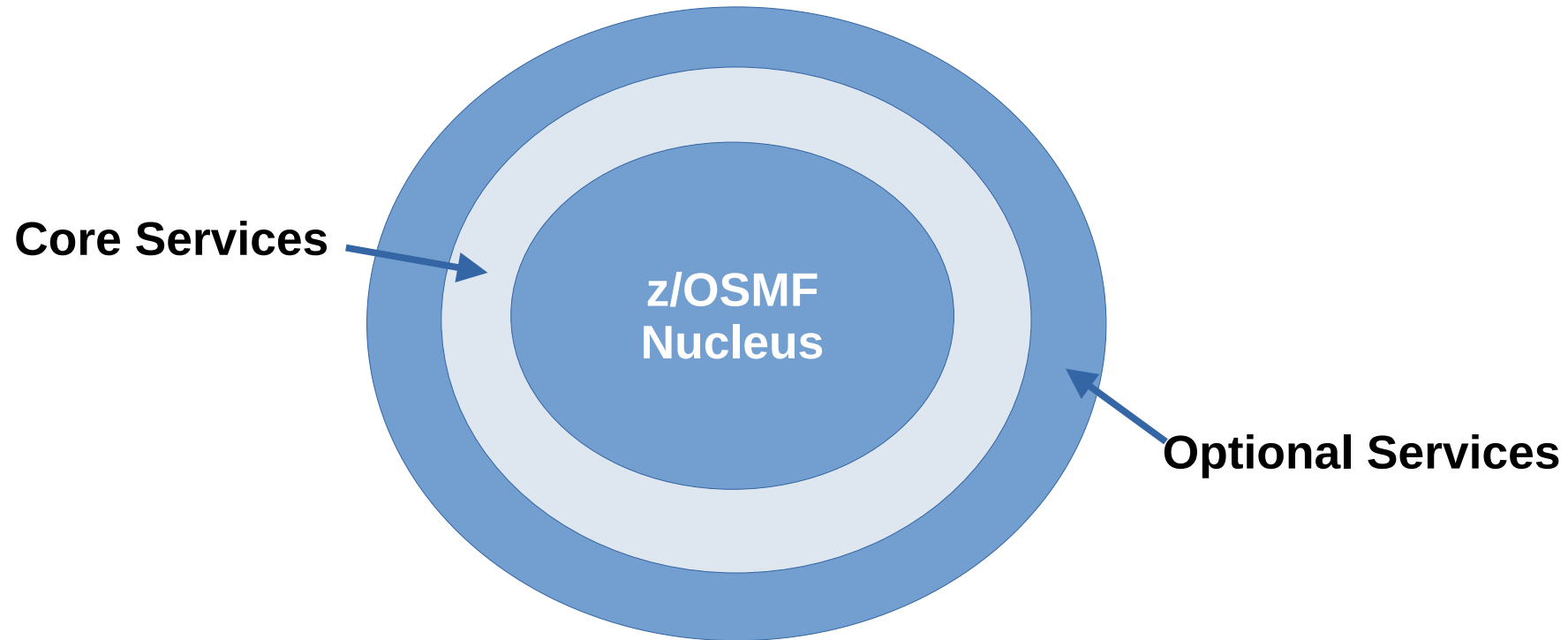
An external tool that can interact with z/OSMF via ssh.

We are here



Z/OSMF OVERVIEW

The z/OS Management Facility is a base component of z/OS and ships with the operating system. It has a bare-bones configuration by default. Installations that intend to make serious use of it configure it further.



Z/OSMF CORE FUNCTIONS

z/OSMF comes with these functions:

- Welcome task
- Notifications and Notification Settings tasks
- Workflows task
- Application Linking Manager task
- Import Manager task
- Links task
- FTP Servers task
- Resource Management task
- Software Services task
- Systems task
- Usage Statistics task
- The z/OSMF online help system.



Z/OSMF PLUG-INS



Optional plug-ins for z/OSMF include:

- Capacity Provisioning
- Configuration Assistant
- Incident Log
- ISPF
- Resource Monitoring
- Software Deployment
- Workload Management

WORKFLOW AUTOMATION

The *workflow* piece of the z/OS Management Facility belongs to a category of software solutions known as *workflow automation* systems. To understand how it works and who uses it, it may be useful to review concepts like *workflow* and *automation*.

While this may sound too basic to warrant a review, keeping the larger context in mind may help you understand why a large organization might want to employ such a tool, and how they may see it playing into their plans for future development of the corporate IT infrastructure.



WORKFLOW



A *workflow* is a series of steps carried out to achieve a goal.

Well, sure. So what?

It isn't just any old series of steps. It's a repeating process that

- requires deep expertise to develop;
- doesn't require deep expertise to follow;
- has to follow company standards and/or legal requirements; and
- must track each step and provide an audit trail.

A workflow usually pertains to an important part of business operations and must be done correctly, and yet consists of individual steps that are tedious, detail-oriented, and sometimes mind-numbingly boring. Those who understand what the workflow does and how to assess its outcome can add more value to the company by spending their time on other things.

MORTGAGE LOAN ORIGINATION WORKFLOW

The process of mortgage loan origination is an example of a workflow common in the financial industry.

The example doesn't pertain to z/OS system maintenance, but is provided to offer context for what a workflow is and why it can be beneficial to automate it.

The next few slides list the steps in a typical mortgage loan origination process.

Don't worry, we aren't going to read through all the steps. The purpose is to illustrate a process that must be performed according to law and company standards without exception, and yet can be handled by personnel with minimal training, provided the workflow itself has been designed by knowledgeable professionals.



MORTGAGE LOAN ORIGINATION WORKFLOW

1. Pre-qualification (optional)

- Borrower provides basic financial information (income, assets, debts, credit estimate).
- Lender gives an approximate idea of how much the borrower may be able to borrow.

2. Pre-approval

- Borrower submits more detailed information, such as:
 - Income documentation
 - Employment history
 - Asset statements
 - Credit information
- Lender performs a preliminary credit review and issues a pre-approval letter indicating a likely loan amount and terms.

MORTGAGE LOAN ORIGINATION WORKFLOW

3. Mortgage application submission

- Borrower selects a property (or applies for refinancing) and completes a formal mortgage application, commonly using the Uniform Residential Loan Application (URLA).
- Required disclosures are provided, including estimated loan terms, costs, and borrower rights.

4. Loan disclosure and document collection

- Lender provides required disclosures and collects supporting documents, such as:
 - Pay stubs
 - Tax returns
 - Bank statements
 - Identification documents
 - Purchase contract (for home purchases)
 - Insurance information

MORTGAGE LOAN ORIGINATION WORKFLOW

5. Loan processing

- A loan processor reviews the file for completeness and accuracy. The processor may:
 - Verify employment and income
 - Order credit reports
 - Confirm assets
 - Request missing documents
 - Prepare the file for underwriting

6. Property appraisal and inspections

- The lender orders an appraisal to determine the property's market value. Other evaluations may occur, depending on the loan and property:
 - Home inspection
 - Pest inspection
 - Environmental reviews
 - Title review

MORTGAGE LOAN ORIGINATION WORKFLOW

7. Underwriting

- A mortgage underwriter evaluates whether the loan meets lending guidelines. The underwriter reviews:
 - Borrower creditworthiness
 - Income stability
 - Debt-to-income ratio
 - Assets and reserves
 - Property value
 - Loan risk
- The loan may receive:
 - Approval
 - Conditional approval (additional items required)
 - Suspension (more information needed)
 - Denial

MORTGAGE LOAN ORIGINATION WORKFLOW

8. Conditions and final approval

- Borrower provides any requested documentation or explanations.

Examples:

- Updated bank statements
 - Proof of deposits
 - Employment verification
 - Explanations for credit issues
- Once conditions are satisfied, the lender issues final approval.

9. Closing preparation

- Lender prepares final loan documents.
- Borrower receives the Closing Disclosure outlining final loan terms and costs.
- Title company or closing agent coordinates signing.

MORTGAGE LOAN ORIGINATION WORKFLOW

10. Closing (loan signing)

- Borrower signs mortgage documents, including:
 - Promissory note
 - Mortgage or deed of trust
 - Closing documents
- Borrower provides required funds for down payment and closing costs.

11. Funding and recording

- Lender releases loan funds.
- The transaction is completed.
- The mortgage lien is recorded with the appropriate local authority.

MORTGAGE LOAN ORIGINATION WORKFLOW

12. Post-closing and servicing transfer (if applicable)

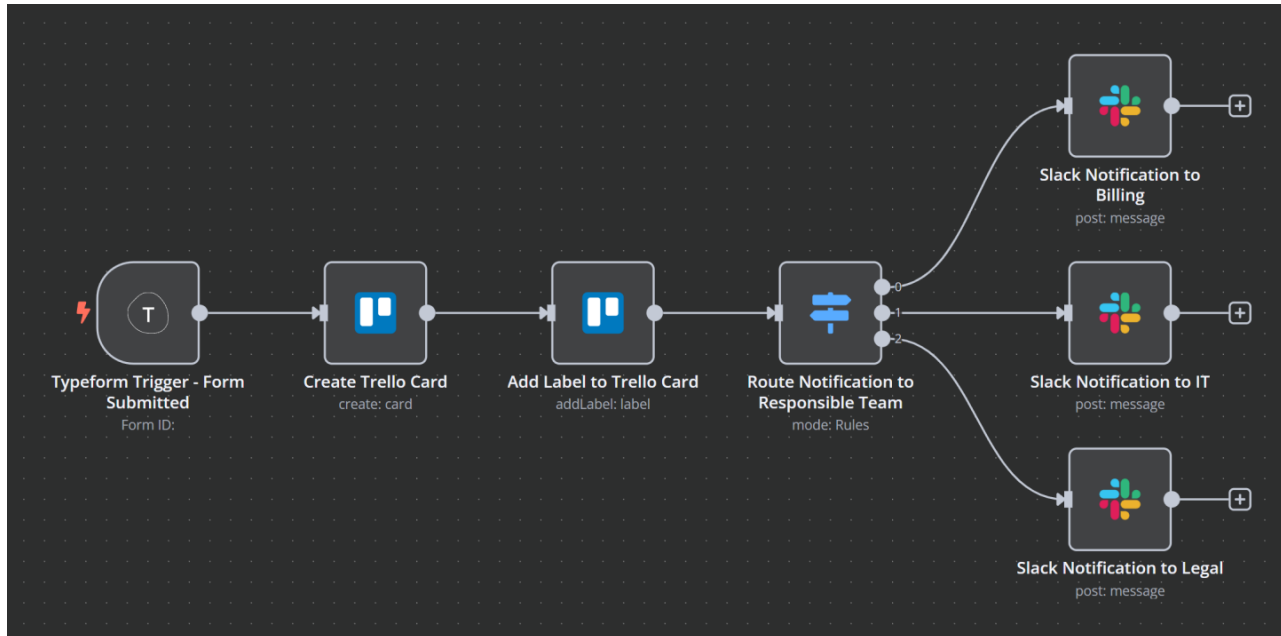
- Loan documents are reviewed and archived.
- The lender may transfer servicing rights to another company.
- Borrower begins making monthly mortgage payments.

WORKFLOW CHARACTERISTICS

Some characteristics of a mortgage loan origination process are similar to those of a process such as installing a software upgrade or onboarding new software developers in a z/OS environment.

- It takes substantial expertise, patience, and effort to work out the details of the process.
- Similarly, it take expertise to understand whether the steps in the process have completed properly.
- Every time the process is carried out, it must be done in the same way, according to company standards, and in compliance with regulations. Any mistakes can "break" the z/OS system and/or result in legal penalties for the company.
- There has to be an audit trail of everything that happens in the process.
- Once the details have been worked out, "anyone" can follow a documented sequence of steps to complete the process (for some definition of "anyone").
- Many of the steps in the process involve coordinating work from multiple organizations, people, and software systems.
- Some of the steps in the process can be performed by software, and need not be carried out manually once the software has been programmed ("automated" steps).
- When regulations or company policies change, the process must be changed in a way that doesn't cause errors or issues. There has to be a controlled way of changing the process.

WORKFLOW AUTOMATION SOFTWARE



Traditional workflow automation software of the kind that might be used to guide workers through a mortgage loan origination process gives the user a list of tasks that must be performed in sequence, and provides a way to track the completion status of each task.

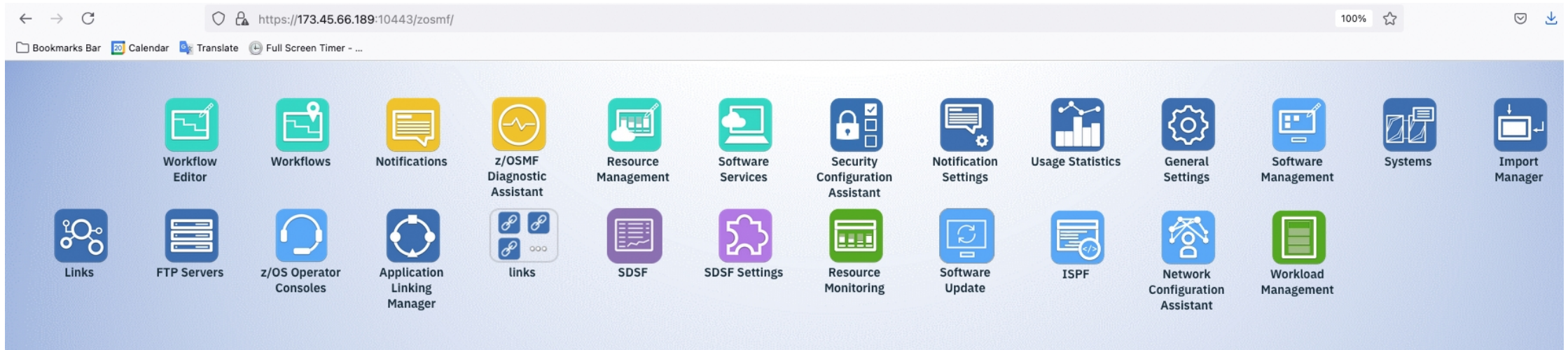
Defining the workflow is a job for people who have deep expertise in mortgage lending and awareness of the various legal and operational requirements for loan origination at their company.

Following the workflow, with guidance from the workflow automation software, is a job for people who have general knowledge of the domain and relatively basic training in using the software.

The screenshot from n8n, a workflow automation product, shows several steps in a sample workflow. The software guides users in carrying out each step in the correct sequence and checking the results.

z/OSMF is in the same category of software solutions. It is specifically designed to support z/OS system administration activities.

WORKFLOW AUTOMATION SOFTWARE



Take a look at the user interface for z/OSMF. It provides screens that guide people in using various z/OS system facilities, such as Resource Management, Security, Software Management, Operator Consoles, Software Update, Workload Management, Network Configuration, and SDSF.

Each of those options guides the user through the process of using the particular z/OS facility. z/OSMF also supports a general-purpose workflow automation feature called, appropriately enough, *Workflows*. The icons for Workflow Editor and Workflows are at the upper left part of the screenshot above.

Z/OSMF OVERVIEW

z/OS configuration and maintenance usually requires running batch jobs and/or entering operator commands on the system, and often requires changes in more than one component using more than one tool.

The z/OSMF web app groups these functions by task, making it easier to perform system management activities that span multiple back-end products or components.

It also can generate jobstreams and sequences of operator commands, relieving the tedium of setting up jobstreams and coding commands, and helping avoid common issues like syntax errors and overlooked or forgotten details of arcane commands.

```
listuser ibmuser
USER=IBMUSER NAME=::::::::::
DEFAULT-GROUP=SYS1 PASSDATE
ATTRIBUTES=SPECIAL OPERATIONS
ATTRIBUTES=PASSPHRASE
REVOKE DATE=NONE RESUME DATE=
LAST-ACCESS=26.186/06:52:32
CLASS AUTHORIZATIONS=NONE
NO-INSTALLATION-DATA
NO-MODEL-NAME
LOGON ALLOWED (DAYS)
-----
AN
G
```

```
//IBMUSERV JOB 'VERIFY SANDBOX CSI',
//          CLASS=A,MSGCLASS=X,MSGLEVEL=(1,1),
//          REGION=0M,
//          NOTIFY=&SYSUID
// EXPORT SYMLIST=(HLQ)
// SET HLQ=PLAT01.CHUMP
//VERIFY EXEC PGM=GIMSMP
//SMPCSI DD DSN=&HLQ..GLOBAL.CSI,DISP=SHR
//SMPLOG DD SYSOUT=*
//SMPOUT DD SYSOUT=*
//SMRPT DD SYSOUT=*
//SMPLIST DD SYSOUT=*
//SMPPTS DD DSN=&HLQ..SMPPTS,DISP=SHR
//SYSPRINT DD SYSOUT=*
//SMPCNTL DD *,SYMBOLS=JCLONLY
SET BDY(GLOBAL). /* List the contents of the Global zone */
LIST.
SET BDY(TARGET). /* ...and the Target zone */
LIST.
SET BDY(DLIB). /* ...and the DLIB zone */
```

```
RESPONSE=VS01
CNZ410SI 06.57.25 DISPLAY ACTIVITY 921
JOBS      M/S      TS USERS      SYSAS      INITS      ACTIVE/MAX VTAM
00018     00033     00001     00034     00031     00001/03500
CNMPSSI   CNMPSSI   NETVIEW   NSW S     CNMPROC   CNMPROC   NETVIEW
RACF      RACF      RACF      NSW S     RRS       RRS       RRS
VLF       VLF       VLF      NSW S     LLA       LLA       LLA
JES2      JES2      IEFPROC   NSW S     CSF       CSF       CSF
SDSF      SDSF      SDSF      NSW S     NET       NET       NET
```

AUTOMATION

The word *automation* in *workflow automation* may be bothering you. Clearly, workflow automation software doesn't make the workflow "automatic." It's a user-facing tool. Some steps might be scriptable or programmable, but on the whole the software is guiding a human through a series of manual tasks. In what sense is that "automated?"

People have different working definitions of *automation* depending on the kind of work they do and on how "automatic" they feel a process must be in order to be labeled, *automated*.

- **Workflow automation.** Software "automatically" guides a human user through a series of steps. It's "automated" compared to the situation where a manager has to tell workers what to do next, again and again every time they carry out the process. With the advent of AI agents, more parts of a workflow may be handled without human intervention than in the past.
- **Executable.** A script or program or a combination of scripts and programs carries out one or more tasks in a series "automatically," once a human starts it. Many people in the IT field refer to anything that is executable as "automated." But the executable Thing still requires a human to start it. It isn't automatic.
- **Automated** in the full sense. A *triggering event* initiates a series of executable steps that can run to completion (or until a "fatal" error occurs) without any human intervention at all. An example is a continuous delivery pipeline that is initiated in response to a push to the main branch of a version control repository. No human has to start the process explicitly.

AUTOMATION

Manual

Sign on to multiple servers and run the same series of commands on each one.

Executable

Write a script to run the series of commands. The script connects to each server and runs the commands. You have to enter certain information manually or reply to prompts or modify environment variables or values in files before starting the script, such as userids, IP addresses, etc.

Runs without human intervention

Change the script so that you don't have to enter values manually or change them in files.

Starts automatically

Configure some sort of monitoring software to initiate the script when something happens in the environment (a *triggering event*). A common example is a CI/CD pipeline that starts whenever someone pushes to main on a Git server.

Z/OSMF AUTOMATED AND MANUAL STEPS

Workflows ▶ Sample demonstrating the use of automated steps in workflow. - Workflow

Sample demonstrating the use of automated steps in workflow.

▶ Workflow Details

Workflow Steps

Actions ▾

↕ No filter applied

☐	State Filter	No. Filter	Title Filter
☐	Unassigned	1	■ Simple automatable step
☐	Unassigned	2	+ Root Step
☐	Unassigned	3	■ Non-automated Step
☐	Unassigned	4	■ Submit an empty JCL job
☐	Unassigned	5	■ Verify the UID of the user using a shell script
☐	Unassigned	6	■ Define the RACF group using a JCL job

z/OSMF Workflows can consist of manual and/or automated steps. This partial screenshot from the z/OSMF UI shows a workflow that has been opened so the user can review its steps.

Some of the steps consist of instructions for the user; those are called *manual* steps. Manual steps can display a form where users can enter information needed to continue the workflow, or they can just tell the user to go and do something.

Other steps consist of references to scripts or jobstreams; those are called *automated* steps. They are not really automated in the full sense of the word, because they don't start automatically in response to a triggering event, without any human intervention.

The user can select which steps they want to apply to their present task, and z/OSMF will guide them through the process of completing those steps.

Z/OSMF WORKFLOWS

Workflows

Settings | Help

Workflows

Simplifies tasks through guided step-based workflows, and provides administrative functions for assigning workflow responsibilities and tracking progress.

Actions ▾ Active ▾ Search

No filter applied

☐ Workflow Name Filter	Description Filter	Version Filter	Vendor Filter	Access Filter	Owner Filter	System Filter
☐ Sample that demonstrates how to run an executable program from a step. - Workflow_OICanceled1782746865739	Sample that demonstrates how to run an executable program from a step.	1.0	IBM	Public	ibmuser	LOCAL.VS01 (
☐ Default Workflow Name	A simple workflow with only one step	1.0	IBM	Public	ibmuser	LOCAL.VS01 (

Total: 2 Selected

Refresh

Workflow Steps

Actions ▾

No filter applied

☐ State Filter	No. Filter	Title Filter
☐ Unassigned	1	A step that runs a UNIX shell script.
☐ Unassigned	2	A step that runs a UNIX REXX exec program.
☐ Unassigned	3	A step that runs a REXX exec program.

Total: 3 Selected: 0

Workflow definitions are coded in XML and must be loaded into z/OSMF before they can be performed. On the z/OSMF UI, this is called *creating* the workflow.

It's a little counterintuitive because you can easily think that "creating" the workflow means coding the workflow definition; but IBM distinguishes between the workflow definition and the workflow itself, which is a software entity running under the control of z/OSMF.

You can think of the workflow definition as the script of a play. When the user "creates" the workflow, it's like staging the play – designing the set, choosing which scenes to cut, changing the dialogue. When the user initiates the first step in the workflow, it's like performing the play.

Z/OSMF ON THE LAB SYSTEM – LOCAL URLS

z/OSMF Main Page

<https://173.45.66.189:10443/zosmf/>

z/OSMF Documentation on site

<https://173.45.66.189:10443/zosmf/helps/>

z/OSMF API Swagger-like documentation

https://173.45.66.189:10443/zosmf/api/explorer/#!/Software_Management_APIs/ListPswis

z/OSMF REST examples

<https://173.45.66.189:10443/zosmf/restfiles/ds/IBMUSER.TEST>

Z/OSMF ON THE LAB SYSTEM – PUBLIC URLS

Overview of z/OSMF in IBM z/OS documentation

<https://www.ibm.com/docs/en/zos/2.3.0?topic=introduction-overview-zosmf>

z/OSMF One Stop Shop

<https://ibm.github.io/zOSMF/>

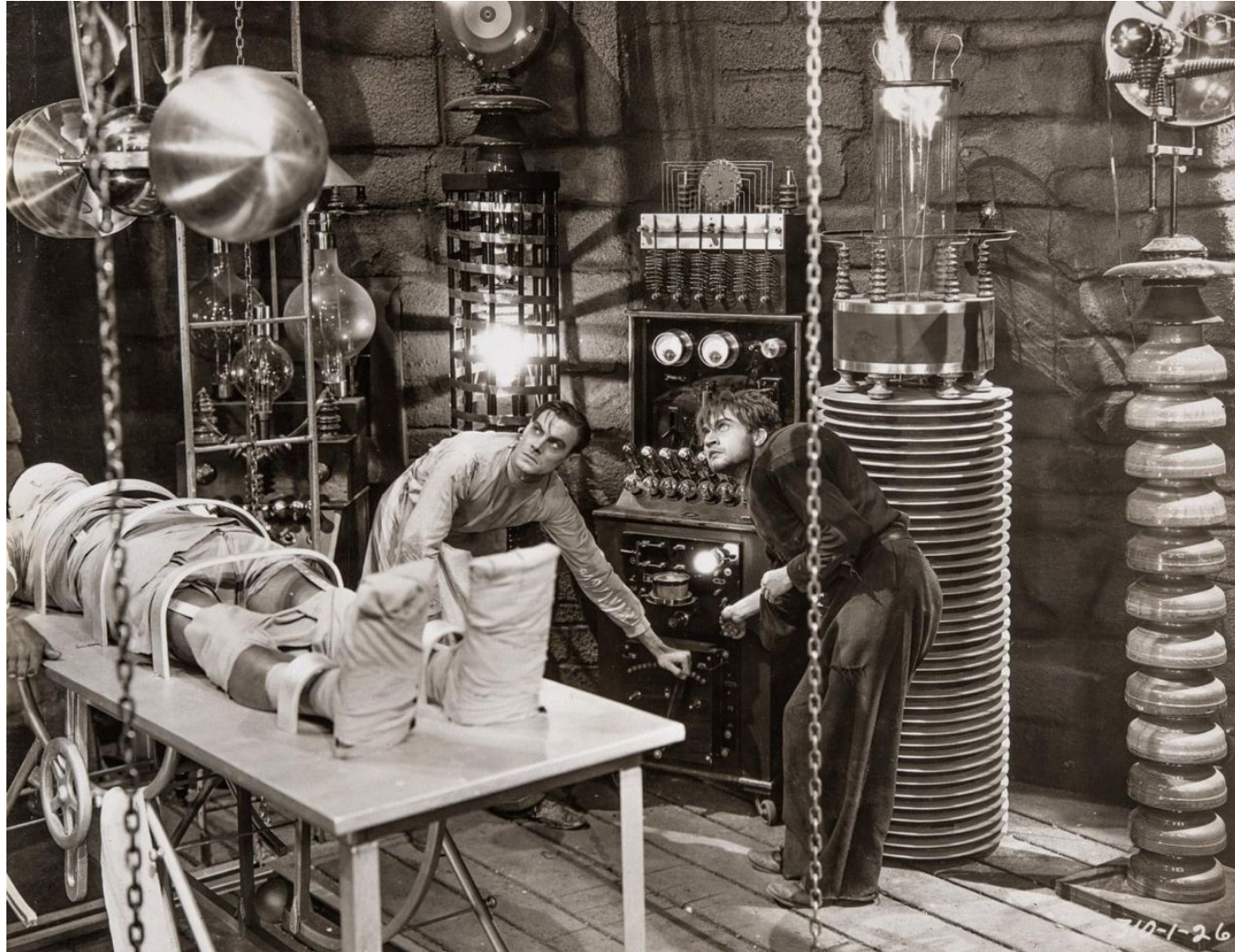
IBM Developer search for "zosmf"

<https://www.ibm.com/docs/en/search/zosmf?type=ibmdeveloper>

List of sample z/OSMF workflows provided by IBM

<https://www.ibm.com/docs/en/zos/3.1.0?topic=file-sample-xml-files-your-reference>

LAB Z/OS 38: EXPLORE Z/OSMF APIS AND WEB INTERFACE



Z/OSMF ON THE LAB SYSTEM - APIS

<https://173.45.66.189:10443/zosmf/api/explorer/#/>

Z/OSMF ON THE LAB SYSTEM – WEB INTERFACE

<https://173.45.66.189:10443/zosmf/>

Z/OSMF ON THE LAB SYSTEM – SAMPLE WORKFLOWS

```
/usr/lpp/zosmf/samples/
```

A WORKFLOW FOR WORKFLOWS



≈UTF-8

IBM provides sample workflows, and we find them on USS. It stands to reason we could copy and modify the samples as a way to get started with building our own workflows. We can do that, in a way, but not by copying and pasting. We have to do it by reading and typing.

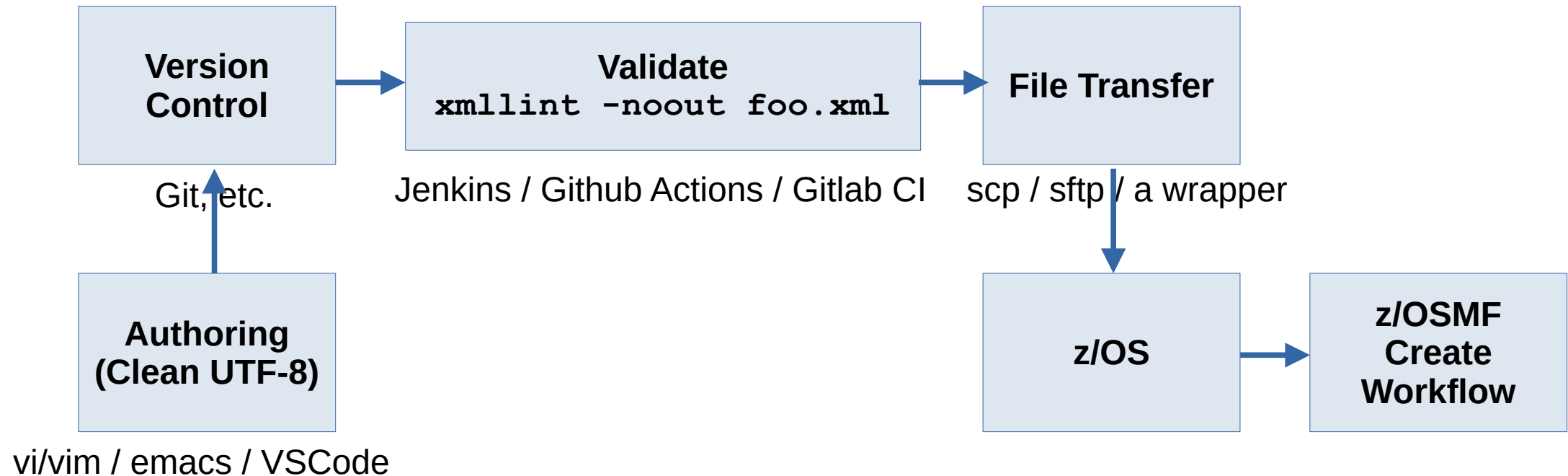
IBM utilities aren't strict about UTF-8 encoding. They're close enough that they can work *with each other's output*, but not close enough that their output works seamlessly with off-platform tools that expect UTF-8 to be *clean* UTF-8. By the time the sample workflows have made it to the samples directory on USS, they've been shoved around quite a bit by various z/OS-based tools.

If you copy them to your own workstation and try to edit them, you'll be disappointed. If you try to edit them on-platform using vi/vim on USS or ISPF Edit on MVS, you'll be disappointed.

This isn't an error on IBM's part. The intent is that the z/OS system should be a target for deploying workflow definitions, and not a platform for creating and editing them. z/OS can *consume* clean UTF-8 files created on another platform.

A WORKFLOW FOR WORKFLOWS

Create and edit z/OSMF workflow definitions off-platform using software development tooling for which XML and UTF-8 are normal. z/OSMF on z/OS will work with well-formed XML in UTF-8 encoding even if it cannot produce the same.



A WORKFLOW FOR WORKFLOWS

Here is a sample directory layout for a z/OSMF workflow authoring environment, assuming something like Linux or Unix.

```
zosmf-workflows/  
  apply-sysmod/  
    apply-sysmod.xml  
    variables.properties  
    steps/  
      step1.xml  
      step2.xml  
  create-racf-userid/  
    create-racf-userid.xml  
    variables.properties  
    step1.xml  
    step2.xml  
  provision-libraries/  
    provision-libraries.xml  
    variables.properties  
    steps/  
      step1.xml  
      step2.xml
```


TWO KINDS OF WORKFLOWS

People talk about "automated" and "manual" workflows, but in fact automation happens at the *step* level, not at the *workflow* level. A workflow can contain both automated and manual steps. A step that contains a script or some other kind of executable is performed automatically by z/OSMF when it is ready to run. The XML code for it might look something like this:

```
<step id="STEP003"
      name="Create user data set">

  <script>
    <type>REXX</type>
    <location>/var/zosmf/workflows/allocate.rexx</location>

    <variables>
      <variable name="HLQ">${USER_HLQ}</variable>
      <variable name="VOLSER">${TARGET_VOLUME}</variable>
    </variables>
  </script>

</step>
```

TWO KINDS OF WORKFLOWS

A manual step only contains instructions for the user, like this:

```
<step id="STEP001"
      name="Verify TCP/IP configuration">

  <description>
    Review the TCP/IP profile and verify that all required
    statements are present.
  </description>

  <instructions>
    1. Open SYS1.TCPPARM(TCPIP) .
    2. Verify the HOME statement.
    3. Confirm the PORT reservations.
    4. Mark this step complete.
  </instructions>

</step>
```

GETTING STARTED WITH THE Z/OSMF UI

This is an IBM training video that walks you through the process of creating a workflow and running it, using the z/OSMF HTTP application.

`https://mediacenter.ibm.com/media/1\_x4v2n1bq`

You can follow along on your company's system or on the class lab system.

WHAT YOU KNOW AND WHAT YOU CAN DO

- You can access the z/OSMF web interface.
- You are somewhat familiar with the screens z/OSMF presents and their functions.
- You can access the live API documentation.
- You are somewhat familiar with the available APIs and their functions.
- You are aware of some of the key limitations of z/OSMF, and that it does not fully eliminate the need for on-platform configuration and maintenance work.

WHAT'S NEXT

- Using Ansible with z/OS managed nodes